Habitation Blocks: System Design Description

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. System overview

The habitation blocks house the resident crew and their families: quarters, mess halls, the medical centre, schools and recreation decks.

The system is led by Life Support & Habitation. It is delivered through 7 work packages with a combined budget of 82,301.03 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
HA-CRQ	Crew quarters module	SHV-HA-SPEC-0003	SHV-HA-DWG-0201	SHV-HA-BOM-0301
HA-MSH	Mess hall	SHV-HA-SPEC-0102	SHV-HA-DWG-0202	SHV-HA-BOM-0302
HA-MED	Medical centre	SHV-HA-SPEC-0103	SHV-HA-DWG-0203	SHV-HA-BOM-0303
HA-SCH	School and childcare wing	SHV-HA-SPEC-0104	SHV-HA-DWG-0204	SHV-HA-BOM-0304
HA-REC	Recreation deck	SHV-HA-SPEC-0105	SHV-HA-DWG-0205	SHV-HA-BOM-0305

3. Architecture

3.1 Crew quarters module

The crew quarters module is specified in SHV-HA-SPEC-0003. Its governing parameter is the berths, set at 2,050 ea. The responsible engineer is Ilka Warrick.

3.2 Mess hall

The mess hall is specified in SHV-HA-SPEC-0102. Its governing parameter is the seating, set at 871 ea. The responsible engineer is Gessa Fiore.

3.3 Medical centre

The medical centre is specified in SHV-HA-SPEC-0103. Its governing parameter is the beds, set at 161 ea. The responsible engineer is Fallon Starke.

3.4 School and childcare wing

The school and childcare wing is specified in SHV-HA-SPEC-0104. Its governing parameter is the classrooms, set at 31 ea. The responsible engineer is Callum Yarrow.

3.5 Recreation deck

The recreation deck is specified in SHV-HA-SPEC-0105. Its governing parameter is the deck area, set at 15,705 m². The responsible engineer is Bale Pike.

4. Interfaces

Document	Title
SHV-ICD-LS-HA	Interface Control Document: Life Support & Atmospherics / Habitation Blocks
SHV-ICD-HA-AG	Interface Control Document: Habitation Blocks / Hydroponics & Food Production

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-HA-01	Habitation Blocks: Design & analysis	Gessa Fiore	COMPLETE	100.0%
WP-SHV-HA-02	Habitation Blocks: Fabricate & install crew quarters module	Sola Quarrel	COMPLETE	100.0%
WP-SHV-HA-03	Habitation Blocks: Fabricate & install mess hall	Treva Bresk	IN_PROGRESS	98.6%
WP-SHV-HA-04	Habitation Blocks: Fabricate & install medical centre	Noa Venner	IN_PROGRESS	80.5%
WP-SHV-HA-05	Habitation Blocks: Fabricate & install school and childcare wing	Ines Everhart	IN_PROGRESS	84.7%
WP-SHV-HA-06	Habitation Blocks: Fabricate & install recreation deck	Sola Quarrel	IN_PROGRESS	79.4%
WP-SHV-HA-07	Habitation Blocks: Integration & test	Brisk Castellan	IN_PROGRESS	53.8%

6. Risks

- Supply of crew quarters module assemblies depends on a single supplier.
- Supply of mess hall assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.